



USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester B.Tech (CBCS) Degree Examinations

December 2024

BT1102-1 – EMERGING BIOPRODUCTS

(BT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which type of diabetes is characterized by insufficient insulin production by the pancreas?
A) Type 1 Diabetes
B) Type 2 Diabetes
C) Gestational Diabetes
D) Autoimmune diabetes
2. Which one of the following is the rich source for SCP
A) E-coli
B) Penicillium
C) Milk protein
D) Spirulina
3. Insulin is belonging to which of the following biomolecule?
A) Carbohydrate
B) Lipid
C) Protein
D) Nucleic acid
4. Which of the following is a broad-spectrum antibiotic?
A) Penicillin G
B) Amoxicillin
C) Clindamycin
D) Daptomycin
5. Bioburanol can be prepared from _____ cell
A) Plant
B) Animal
C) Microbial
D) Insect
6. What is the primary advantage of using subunit vaccines?
A) They reduce the risk of adverse reactions.
B) They include live pathogens to elicit immunity.
C) They require no refrigeration.
D) They are less expensive to produce than whole pathogen vaccines.
7. What are bioplastics primarily made from?
A) Fossil fuels
B) Recycled plastics
C) Renewable biological resources
D) Synthetic chemicals
8. Which microbial-based vaccine is commonly used to prevent cervical cancer?
A) Influenza vaccine
B) HPV vaccine
C) Hepatitis B vaccine
D) MMR vaccine
9. Which of the following is a common type of bioplastic?
A) Polyethylene
B) Polylactic acid
C) Polyvinyl chloride
D) polypropylene
10. Pluripotent stem cells refer to _____
A) Plural form stem cell
B) Can differentiate to one type of cells
C) Can differentiate to any type of cells
D) Polyform of stem cells

11. One of the main advantages of bioplastics is _____
 A) They are cheaper to produce than traditional plastics.
 B) They decompose more rapidly than conventional plastics.
 C) They have a higher strength-to-weight ratio.
 D) They are made entirely from petroleum.
12. What is the primary function of digestive enzyme supplements?
 A) To increase muscle mass
 B) To enhance energy levels
 C) To aid in the breakdown of food
 D) To support immune function
13. Which of the following is a potential benefit of using single-cell protein in food production?
 A) High cost of production
 B) Limited nutritional value
 C) Sustainability and efficient use of resources
 D) Requires extensive land use
14. Which enzyme-based product is frequently used in the brewing process to break down starches into fermentable sugars?
 A) Protease
 B) Amylase
 C) Cellulase
 D) Lactase
15. What is one of the primary applications of stem cell therapeutics in medicine?
 A) Cosmetic enhancements
 B) Gene editing
 C) Regenerative medicine
 D) Vaccine development
16. Which microorganism is commonly used to produce single-cell protein for animal feed?
 A) *Saccharomyces cerevisiae*
 B) *Escherichia coli*
 C) *Streptomyces* spp.
 D) *Lactobacillus* spp.
17. Which of the following is a common application of monoclonal antibodies?
 A) Treating viral infections
 B) Cancer immunotherapy
 C) Pain relief
 D) Hormone replacement therapy
18. Which of the following is a well-known example of a recombinant protein used in medicine?
 A) Hemoglobin
 B) Collagen
 C) Insulin
 D) Albumin
19. What is a major advantage of using natural dyes over synthetic dyes?
 A) Natural dyes are more vibrant.
 B) Natural dyes are less expensive to produce.
 C) Natural dyes are more resistant to fading.
 D) Natural dyes are biodegradable
20. Which of the following is a significant bioproduct derived from algae that is commonly used as a dietary supplement?
 A) Chlorella
 B) Spirulina
 C) Agar
 D) All of these

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) How do different types of vaccines (live-attenuated, inactivated, subunit) compare in terms of effectiveness and safety?	08	L*2	1	1
	b) How is bioenergy produced from biomass, and what technologies are commonly used in this conversion process?	08	L1	2	1

- | | | | | | |
|-------|--|----|----|---|---|
| 2. a) | What are the key differences between traditional plastics and bioplastics in terms of composition, production processes, and biodegradability? | 08 | L1 | 1 | 1 |
| b) | What is xanthan gum, and how is it produced through the fermentation process using the bacterium <i>Xanthomonas campestris</i> ? | 08 | L2 | 2 | 1 |
| 3. a) | What are bioconversion products, and how do microorganisms or enzymes transform raw materials into valuable substances in bioprocessing? | 08 | L2 | 2 | 1 |
| b) | What is single-cell protein (SCP), and how is it produced from various microbial sources, such as bacteria, yeast, and fungi? | 08 | L2 | 1 | 1 |

Unit – II

- | | | | | | |
|-------|--|----|----|---|---|
| 4. a) | What are the key differences between biologics and traditional small-molecule drugs in terms of their mechanisms of action, manufacturing processes? | 08 | L1 | 2 | 1 |
| b) | What are the challenges associated with the clinical use of monoclonal antibodies? | 08 | L2 | 4 | 1 |
| 5. a) | What are the advantages of using animal cell cultures to produce vaccines compared to traditional methods? | 08 | L1 | 3 | 1 |
| b) | What are the key steps involved in the discovery, development, and production of therapeutic proteins? | 08 | L1 | 3 | 1 |
| 6. a) | How do recombinant proteins function as therapeutic agents in the treatment of diseases? | 08 | L2 | 4 | 1 |
| b) | Differentiate between embryonic stem cells, adult stem cells. | 08 | L1 | 3 | 1 |

Unit – III

- | | | | | | |
|-------|---|----|----|---|-----|
| 7. a) | How do plant secondary metabolites contribute to human health, particularly in the context of traditional medicine? | 08 | L1 | 5 | 1 |
| b) | What are the potential benefits and limitations of using natural dyes in various applications? | 08 | L1 | 5 | 1 |
| 8. a) | What are the primary types of algal bioproducts, such as biofuels, pigments, and nutraceuticals? | 08 | L1 | 5 | 1,2 |
| b) | Discuss about plant based compounds in cosmetic applications. | 08 | L2 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
